COM3105 Advanced Algorithms

Homework 4

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Question 1

(5 points)

Consider a stream $a_1, a_2, \ldots, a_m$, where each $a_i \in [n]$.
Design and analyze a streaming algorithm that, given a query
$S \subseteq [n]$,
outputs an estimate of what fraction of items in the stream belong to S within additive error ϵ with probability at least δ.
The space used by the algorithm should only depend on ϵ and δ.
Note that S is given only at query time (not in advance).

Answer

The intuition behind my answer is to use reservoir sampling with k samples to estimate the fraction of streamed elements in S.
Algorithm:

- **Initialisation:** Choose a sample size k based on the parameters:

  $$k \geq \frac{1}{4\,\epsilon^2\,\delta}$$

  Where ϵ is the chosen additive error and δ is the probability of failure. Initialise an empty set (the reservoir) R that stores k items. Initialise a counter $C = 0$ to track the number of items seen so far.

- **Processing the stream:** For each incoming element a_i, where i is the index of the element in the stream (starting from 1):

  - If $C < k$, insert a_i into R, and increment C by 1.
  - Otherwise:
    - Include a_i in the reservoir R with probability k/i.
    - If included, replace a uniformly random element from R.
    - Increment C by 1.

  This is standard reservoir sampling.

- **Query:** When the set S is given, compute the estimate:

  $$\widehat{p} = \frac{1}{k} \sum_{x \in R} \mathbf{1}\{x \in S\},$$

  and output $\widehat{p}$ as the estimate of the true fraction.

Expectation of the Estimator:
Let p be the true fraction of stream elements that belong to S:

$$p = \frac{1}{m} \left|\{i : a_i \in S\}\right|$$

After reservoir sampling has finished, the reservoir contains k uniformly random elements from the stream. For each $j = 1, \dots, k$, define the indicator variable

$$X_j = \mathbf{1}\{R_j \in S\},$$

, where R_j is the j-th sampled item.
Because reservoir sampling picks each stream element with equal probability, each R_j is distributed exactly as a uniformly random stream element. So,

$$\mathbb{P}(R_j \in S) = p,$$

and therefore

$$\mathbb{E}[X_j] = 1 \cdot p + 0 \cdot (1-p) = p$$

The estimator for the fraction is

$$\widehat{p} = \frac{1}{k} \sum_{j=1}^{k} X_j$$

Using linearity of expectation,

$$\begin{aligned} \mathbb{E}[\widehat{p}] &= \mathbb{E}\left[\frac{1}{k} \sum_{j=1}^{k} X_j\right] \\ &= \frac{1}{k} \sum_{j=1}^{k} \mathbb{E}[X_j] \\ &= \frac{1}{k} \sum_{j=1}^{k} p \\ &= \frac{1}{k} \cdot kp \\ &= p \end{aligned}$$

So the estimator has expectation equal to the true fraction:

$$\mathbb{E}[\widehat{p}] = p$$

Variance of the Estimator:
Remember that each random variable X_j is an indicator:

$$X_j = \begin{cases} 1 & \text{if the } j\text{-th sampled element belongs to } S, \\ 0 & \text{otherwise} \end{cases}$$

So,

$$\mathbb{P}(X_j = 1) = p \qquad \text{and} \qquad \mathbb{P}(X_j = 0) = 1 - p$$

To get the variance of X_j we can use the formula:

$$\mathrm{Var}(X_j) = \mathbb{E}[X_j^2] - (\mathbb{E}[X_j])^2$$

Since $X_j^2 = X_j$ (as the indicators are either 0 or 1):

$$\mathbb{E}[X_j^2] = \mathbb{E}[X_j]$$

And from the expectation calculation above we have:

$$\mathbb{E}[X_j] = p$$

Putting this together:

$$\mathrm{Var}(X_j) = p - p^2 = p(1 - p).$$

Because $p(1 - p)$ is maximized when $p = \frac{1}{2}$, we know that:

$$\mathrm{Var}(X_j) = p(1 - p) \leq \frac{1}{4}$$

Remembering that the estimator is the mean of the X_j in the reservoir:

$$\widehat{p} = \frac{1}{k} \sum_{j=1}^{k} X_j$$

Remembering the two properties of variance:

- For any random variable Y and constant c,

$$\mathrm{Var}(cY) = c^2 \,\mathrm{Var}(Y)$$

- For independent random variables Y_1, Y_2,

$$\mathrm{Var}(Y_1 + Y_2) = \mathrm{Var}(Y_1) + \mathrm{Var}(Y_2)$$

As the X_j are independent, we can use these properties to compute the variance of $\widehat{p}$:

$$\mathrm{Var}(\widehat{p}) = \mathrm{Var}\left(\frac{1}{k} \sum_{j=1}^{k} X_j\right) = \frac{1}{k^2} \sum_{j=1}^{k} \mathrm{Var}(X_j)$$

Using $\mathrm{Var}(X_j) \leq 1/4$ for every j:

$$\mathrm{Var}(\widehat{p}) \leq \frac{1}{k^2} \cdot k \cdot \frac{1}{4} = \frac{1}{4k}$$

Quality Analysis Using Chebyshev's Inequality:
Chebyshev's inquality is defined as:

$$\mathbb{P}(|X-\mu| > \lambda) \leq \frac{\mathrm{Var}(X)}{\lambda^2}$$

Using Chebyshev's inequality for our estimator $\widehat{p}$, we can substitute in our bounded variance (from above) and set $\lambda = \epsilon$:

$$\mathbb{P}\left(|\widehat{p}-p| > \epsilon\right) \leq \frac{1}{4k\epsilon^2}$$

To bound the probability of error, we set the right-hand side to be at most δ':

$$\frac{1}{4k\epsilon^2} \leq \delta'$$

Now we solve for k:

$$\frac{1}{4\epsilon^2\delta'} \leq k$$

Substituting this into the right-hand side:

$$\mathbb{P}\big(|\widehat{p}-p| > \epsilon\big) \leq \frac{1}{4 \cdot \left(\frac{1}{4\epsilon^2\delta'}\right) \cdot \epsilon^2} = \frac{1}{\frac{1}{\delta'}} = \delta'$$

Therefore, with this choice of k we have:

$$\mathbb{P}\big(|\widehat{p}-p| > \epsilon\big) \leq \delta'$$

Which we can rewrite as:

$$\mathbb{P}\big(|\widehat{p}-p| \leq \epsilon\big) \geq 1-\delta'$$

Now we can set $1-\delta' = \delta$ to bound the probability of error by δ:

$$\mathbb{P}\big(|\widehat{p}-p| \leq \epsilon\big) \geq \delta$$

This proves that the estimator $\widehat{p}$ is within additive error ϵ of the true fraction p with probability at least δ.

Space Complexity Analysis:
The algorithm stores:

- A reservoir of size $k = \mathcal{O}\left(\frac{1}{4\epsilon^2(1-\delta)}\right)$.
- Each stored item in k is from $[n]$ and requires $\mathcal{O}(\log n)$ bits.

Therefore, the total space used by the algorithm is:

$$\mathcal{O}(k\log n) = \mathcal{O}\left(\frac{1}{\epsilon^2(1-\delta)}\log n\right)$$

This space depends only on ϵ and δ (and $\log n$, but this is not tuneable), as required.

Question 2

(5 points)

question here

Answer

answer here